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Monetary sanctions and deterrence: the effect of speeding tickets on road behavior

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Abstract: Governments use monetary fines in order to deter socially unwanted behavior on a wide array of issues. It is commonly assumed that, fixing the probability of sanction, the higher the price, the more deterrent a sanction is. I use data from automated speed cameras in Bogotá, Colombia to measure the deterrent effect of a speeding ticket. By using the real speed limit as a source of exogenous variation and a regression discontinuity design, I find that speeding tickets have a general deterrent effect on speeding behavior but a null effect on reducing traffic accidents. I show that the deterrent effect is concentrated on cars, and absent on motorcycles. Using the commercial price of the vehicle as a proxy for wealth and the fact that the monetary sanction is fixed, I find that the deterrent effect is concentrated on the higher quintiles of the distribution. I validate these findings with a judge-IV using the fact that a police agent manually verifies all tickets. In short, speeding tickets might be less useful for preventing road fatalities that is commonly assumed and inefficiently expensive for a considerable number of drivers. They also seem to be of little effectiveness for the current public health challenge of traffic fatalities.

1. Introduction:

A central question for governments is how to improve the deterrent power of their legal sanctions. A common avenue is to increase the monetary sanctions under the assumption that a higher expected cost results in a lesser prevalence of unwanted behavior. In context with little enforcement this assumption might be wrong; legal sanctions can be too expensive for certain actors that not only avoid paying the fine but also continue with unwanted behavior.

This paper explores the question if a legal monetary sanction can be too expensive. I use traffic tickets and accidents to answer it. I explore the differential effect of relative price on behavior in a context in which there is a fixed price for a speeding infraction, but it can be compared to the market price of the vehicle. The possible

selection of drivers into law breakers and law-abiding drivers is dealt with by using a regression discontinuity design with the unknown gap between the posted street limit and the real speed at which speeding tickets are given.

This article starts by answering questions on the general causal effect of speeding tickets on behavior: What is the causal effect of a speeding ticket on subsequent behavior of a driver? The most studied subsequent behavior usually speeding itself, but with the data available I can also observe the chances of being involved in an accident. Since speeding tickets are used as a tool for preventing road accidents, this is the most relevant variable to study. I will argue that studying the effect of traffic tickets on subsequent speed tickets is like measuring the effect of taking a pill to reduce cholesterol on taking the next pill, not on cholesterol levels.

This paper relates to the literature on financial fines and fees (commonly referred to as legal financial obligations, or LFOs). There is debate on the capacity of higher fines to increase revenue and the effect this has on recidivism. Some papers find a modest effect on revenue at the cost of higher recidivism (Giles, 2023). Others argue that even if the effect of sanctions is higher revenue, they have no effect on recidivism and thus are not justifiable ways of increasing deterrence (Finlay et al., 2024). The authors argue that the null effect in all relevant criminal behaviors is precisely estimated, not to lack of power. In recent work, (Norris & Rose, 2024) argue that reducing LFOs will impact municipal revenues without increasing deterrence. The evidence provided by (Bing et al., 2024) on a randomized payment of existing fines for criminals shows that this policy only reduces recidivism related to the financial costs of the LFO itself. These effects are not negligible and point to no deterrent effect at a significant cost of engagement with the criminal system. In summary, LFO's seem to have a null effect on criminal behavior, while a modest effect on increasing municipal revenue. Questions of fairness arise from these two facts.

This question has also been examined in the specific case of traffic tickets, but mainly through the lens of revenue. Some have shown that increasing fines has only a modest effect on timely payment (Traxler & Dušek, 2023) and recidivism (Dušek & Traxler, 2022). A very detailed of the effect of the price of ticket fines is (Kaila, 2024), who uses the relative to income price of traffic tickets in Finland to show that there is in fact a reduction on recidivism for higher fines, but the effect is short lived and the fines need to be significantly more expensive. In their work, (Goncalves & Mello, 2024) also study the effect of harsher sanctions by exploiting the discretionary sanctions given by police officers. They find that harsher fines do reduce recidivism and, to a lesser degree, crash involvement. They also find that officers discretion is inefficient since harsher sanctions are given to drivers that are less deterred by them. Contributing to this new evidence on the indirect deterrent effect of traffic fines on road safety is one of the central contributions of this paper.

Another branch of the literatures studies what citizens learn by interacting with law enforcement. Using a change on the punishment for recidivism, (Philippe, 2024) finds that learning about criminal law appears to be limited to people with direct involvement with the law. In the same direction, it has been shown that individuals do adjust their priors after receiving a speeding ticket (Dušek & Traxler, 2022). As I will explain later, I contribute to this literature by studying subsequent behavior where there is going to be future interactions with the law and also where there is not. I can speak to how interactions with the law affects behavior when citizens know the law is not present.

Finally, this paper also contributes to a broader literature on the effect of traffic enforcement on behaviors (De Angelo & Hansen, 2014; Hansen, 2015; Luca, 2015; Makowsky & Stratmann, 2011). In particular, my works add to a growing literature that studies fines and enforcement in relation to factors that do not affect either sanctions or probability of enforcement but have been shown to have an effect on traffic accidents. Things like the political cycle (Bertoli & Grembi, 2021), or the stock market (Fry & Farrell, 2023) (Giulietti et al., 2020). Close to this strand is a set of papers that explores behavioral responses to traffic enforcement and tries to derive policy recommendations to reduce road accidents and fatalities (Zhang et al., 2020) (Lu et al., 2016). My work contributes by modelling and measuring perceptions of fairness of sanctions as a mediator of changes in enforcement. I also contribute to this literature by adding evidence from a city in the developing world, something that is still rare in the field.

Examining the effect of speeding tickets on behavior is difficult because of selection. Drivers that decide to exceed speed limits might be significantly different in unobservable characteristics that also relate to risky behavior, thus comparison between ticketed and not ticketed drivers cannot capture the causal effect of tickets. But the real working of the process for giving speed tickets offers a way to solve this issue. Speed tickets are not given to all vehicles that exceed the maximum speed posted on road signs, but to a different speed with an additional tolerance. Since the number of kilometers per hour of this tolerance is not known to drivers, thus a discontinuity exists. A regression discontinuity design will be used to make a valid comparison between only drivers that decide to exceed the speed limit, but some receive a ticket and some do not. Evidence for the validity of the design will be provided on the methodology chapter of this paper.

The paper proceeds as follows. Section 2 presents the institutional background of speed fines in Bogotá, Colombia and the Data that will be used on the estimation. Section 3 discusses the estimation strategy and presents the evidence necessary for its validity. Section 4 presents all the results from the estimation on the general effect of speeding tickets with a fixed price. Section 5 presents results on the relative price of vehicles. Section 6 presents some robustness checks on the main estimation. Section 7 briefly discusses the behavioral mechanisms that could be in

place. Finally, section 8 discusses the main findings in relation to fines as an effective deterrent strategy and derives possible policy recommendations for its improvement.

2. Institutional background and data:

Automated enforcement of speed started in Bogotá in 2020, with the staggered installation of Cameras in the most dangerous zones. The program was originally called *Cámaras Salvavidas* (Spanish for Lifesaving cameras) to emphasize that the objective of the program was not municipal revenue but behavioral change. After some debate on the legality of the fines, Colombia's higher court¹ allowed for the fine to be given to the owner of the vehicle. Two elements of the legal frame are crucial in the rest of the paper. First, tickets have a fixed price independent² of the type of vehicle or the income of the owner, or the speed at which a vehicle is circulating. Second, there is no additional fine for recidivism.

The central database I will use for defining treatment includes all vehicles that exceeded the legal speed limit under a Speed Camera from August 16 to October 15, 2024. This dataset is comprised of 181,489 distinct vehicles that passed the speed Cameras at the time at which they give speeding tickets, which corresponds to daylight. Since I will only be using vehicles that passed under the Cameras once a total of 158,703 comprise the full sample used for estimation³. This is equivalent to 87% of the plates that exceeded the limit. I will call this sample my unmatched sample. On the Appendix section 2 shows some results using also some vehicles that passed more than once.

I can also access the databases for all existing tickets and accidents occurring in Bogotá. I can see if each plate received a ticket or was involved in an accident both before and after the period of study, until March 15 of 2025. Those two will be my main outcome variables but also are important in correcting for specific characteristics of the sample.

I can access the registration record for all vehicles in Bogotá (and some that pay congestion pricing⁴) and match the plate with information on brand, model and year of the vehicle. A web scrapping procedure of the most used website for selling used cars in the city is done to obtain the commercial value of each brand -model-year

¹ This is ruling C-321 of 2022 by Corte Constitucional del Colombia.

² Technically there is an early payment, but all vehicles are subject to this. So the price is the same for everyone.

³ Full distribution of the number of observation by vehicle in this sample is shown on Table 1 of the Appendix. This is a particular problem since vehicles might circulate one time below the real limit and another one above the real limit, thus making it harder to classify them as treated or untreated in terms of the discontinuity. I will do robustness checks on this, but mostly just use vehicles that circulated once.

⁴ Some vehicles can pay to avoid a circulation restriction that exists in the city (*Pico y placa Solidario*). All the data I need is registered on these transactions so I can access the data in this case.

combination⁵. This leaves me with a total of 67,235 vehicles for which I can find the commercial prices, naturally a subsample but is big enough for the estimation to have statistical power. I will use the price of the vehicle as a proxy for wealth. The heterogenous effect of the relative price of the ticket can thus be estimated. Local effects and external validity questions will be analyzed in the Results section. I will call this sample my matched sample.

In no way am I arguing that my matched sample is similar to the Full Sample. The decision to register the vehicle in Bogotá is surely related to behavior as the fact that the registry is growing in surrounding municipalities but not in Bogotá. This is particularly caused by the decision of new motorcycle owners to register their vehicle outside Bogotá, even if this is where they plan to use it. Thus, the effect seen on that sample might only extend to other drivers that decide to register their vehicles in Bogotá.

3. Estimation strategy.

My estimation derives from a fact that is unknown to drivers, that tickets are not given at the speed posted on the road. On the following I will call the legal limit posted on the road sign (50 km/h) as the posted speed limit, and the limit at which most tickets really given (55 km/h) as the real limit⁶.

This discontinuity allows for some exogeneous variation that can be exploited. In most of my applications I will estimate the following equation:

$$Result_i = f(z_i) + \tau D_i + \epsilon_i \quad (1)$$

Where $Result_i$ is one of my three behaviors of interest for vehicle i : a dummy that takes the value of 1 if the vehicle had at least one instance of excess speed⁷ in the six months after receiving a ticket and 0 otherwise, a dummy that takes the value of 1 if the vehicle had one accident in the six months after receiving a ticket and 0 otherwise, or the total number of instances of excess speed in the six months after receiving a ticket.

$f(z_i)$ is a polynomial of degree 2 (Gelman & Imbens, 2019) for the variable that decides treatment, the speed at which vehicle i is circulating under a Speed Camera. All vehicles in the sample exceed the posted speed limit, but only some exceed the real limit. This reduces selection and allows me to identify causality.

D_i is my treatment variable, that takes the value of 1 if the vehicle received a speeding ticket and 0 otherwise. The treatment is instrumented via the assignment,

⁵ All data on vehicle prices is taken from tucarro.com, the most used website for selling used vehicles.

⁶ In the sample there is only one Camera that has 30 as the maximum speed. It only accounts for 4,460 observations (1.82%). Speeds are adjusted accordingly.

⁷ Since the real limit is unknown to drivers, a future excess speed is the one at above the posted limit; that is 50 km/h, not 55 km/h.

since this is a fuzzy design, and the discontinuity at the real speed limit is around an increase of 27 p.p. on the probability of receiving a ticket.

ϵ_i is an error term.

τ is my coefficient of interest. I will argue that it captures the causal effect of receiving a ticket in subsequent behavior. It is a local effect, since it only captures the effect of a ticket for vehicles around the cutoff, but those are the vast majority of observations in the sample⁸. For all estimations I use the `rdrobust` package in STATA with the option to ignore mass points, as the variable that assigns treatment is discrete. Three estimator, conventional, bias-corrected and robust will be shown.

3.1 Validation of the Fuzzy Regression Discontinuity Design: Existence of discontinuity:

Figure 1 shows that enforcement does work with the real limit, not the one posted on the street signs. It also shows that the discontinuity is not 0 to 1, so my strategy will be a fuzzy design. In short, even if the behavior of driving at 55 km/h is very similar to driving at 56 km/h, and both are above the legal limit, the latter has four times the probability of receiving a ticket than the former.

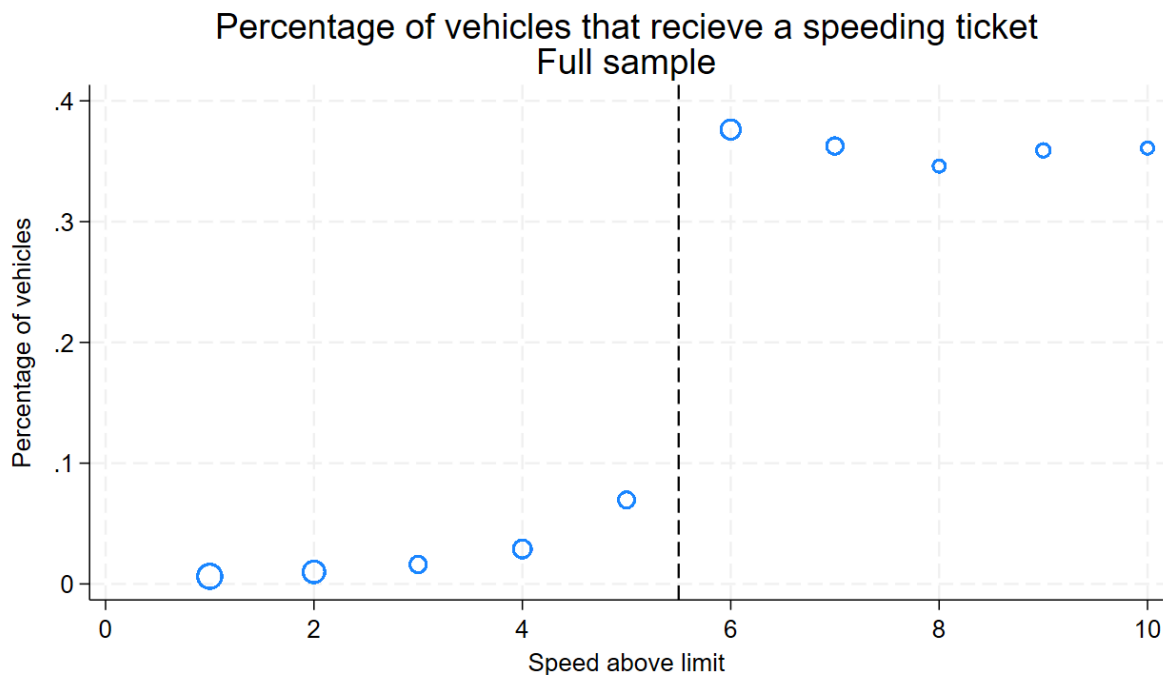


Figure 1. Speed above limit is constructed as: (vehicle observed speed-speed limit).

It is important to show that this discontinuity also exists on the matched sample and that is similar for different values of vehicle prices. Figure 2 plots this discontinuity

⁸ 80% of observations pass under Speed Cameras at speeds between 51 and 60 km/h.

for all matched vehicles and for each quintile of vehicle price. In fact, the discontinuity is slightly bigger in this case across the board, but particularly for higher quintiles.

Percentage of vehicles that receive a speeding ticket By quintile

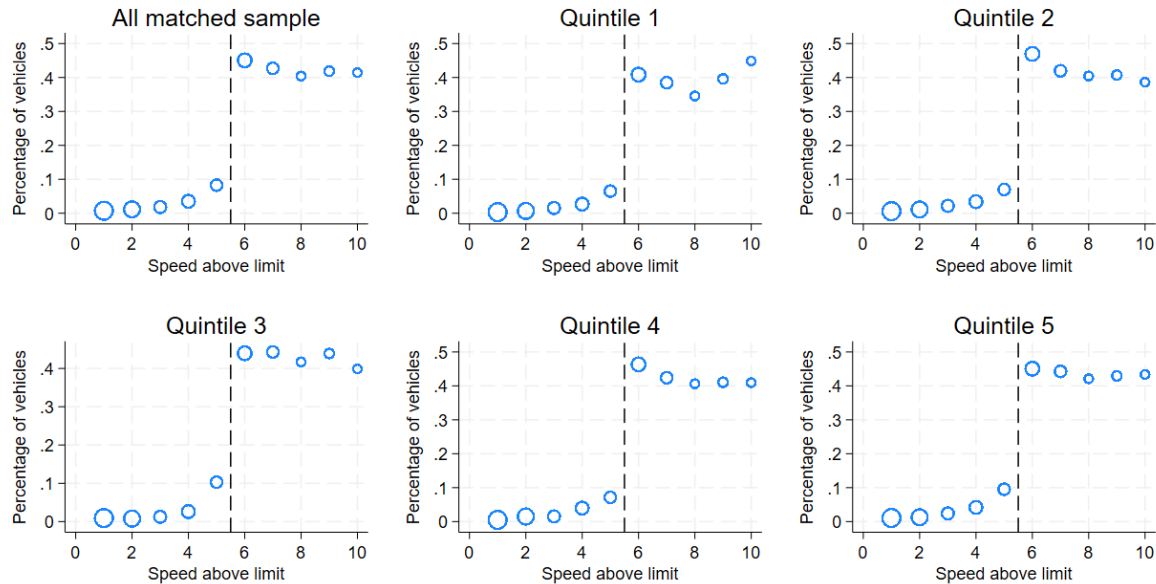
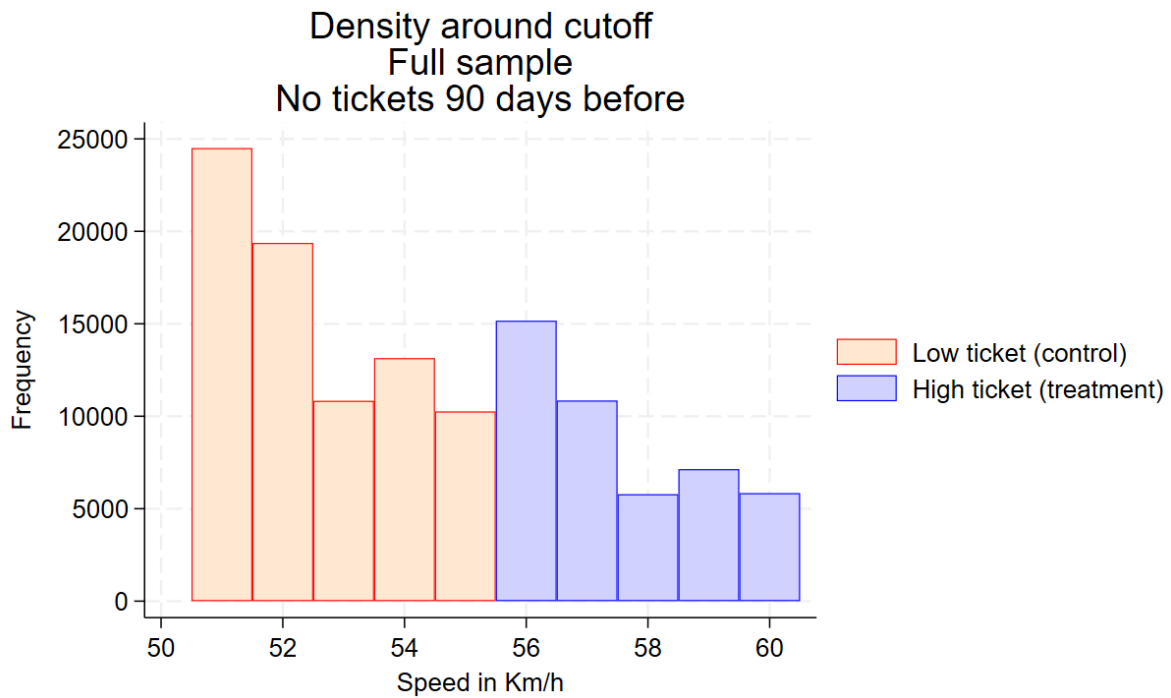


Figure 2. Speed above limit is constructed as: (vehicle observed speed-speed limit).

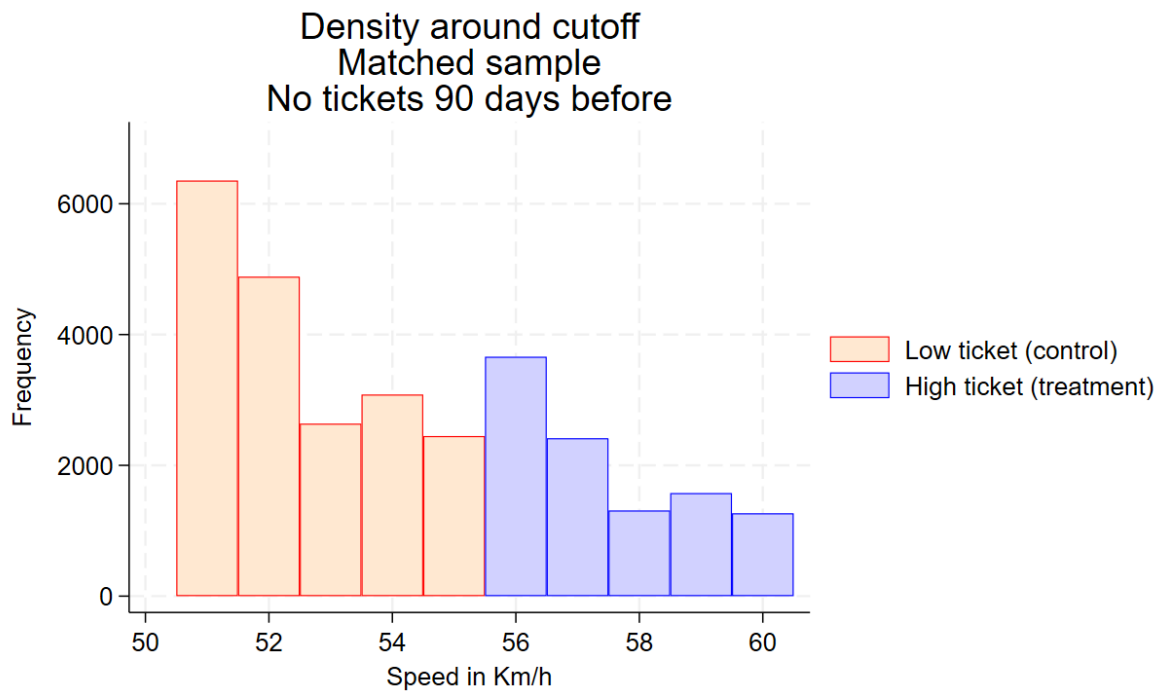
A very relevant question for local public policy is whether this deterrent effect of tickets varies by vehicle type. This is caused by motorcycles being a growing percentage of road injuries and fatalities both in Bogotá and in all Colombian cities, as in similar middle income countries. Graphs 3 and 4 of the Appendix show this for the full and matched sample. It can be seen that the discontinuity is smaller for motorcycles, and that this is considerably smaller in the case of the matched sample. This is less a threat for the identification strategy and more of a fact to have in mind when interpreting results.

3.2 No manipulation.

Now that I have shown that there is a discontinuity at cutoff, I need to show that there is no manipulation. In short, drivers are not aware of this fact. Since the estimation wants to separate the effect of a possible ticket from any other change, I will only use the vehicles that have not received a ticket in the 90 days before passing under a Camera exceeding the legal speed limit. Frequency is shown in Graph 5. In the Appendix the same distribution is shown for all vehicles, independent of receiving or not receiving a ticket in the prior 90 days. It seems that drivers are not selecting themselves into not receiving a ticket.

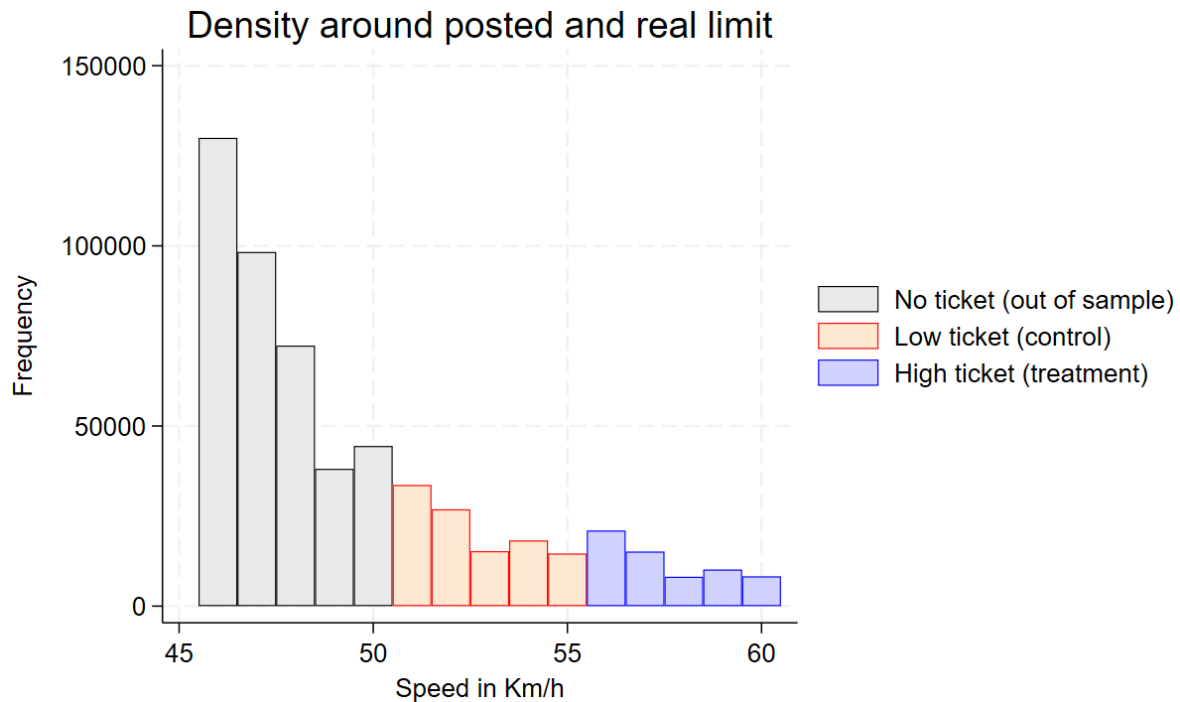


Graph 3. Frequency by condition. Full and sample.



Graph 4. Frequency by condition. Matched sample.

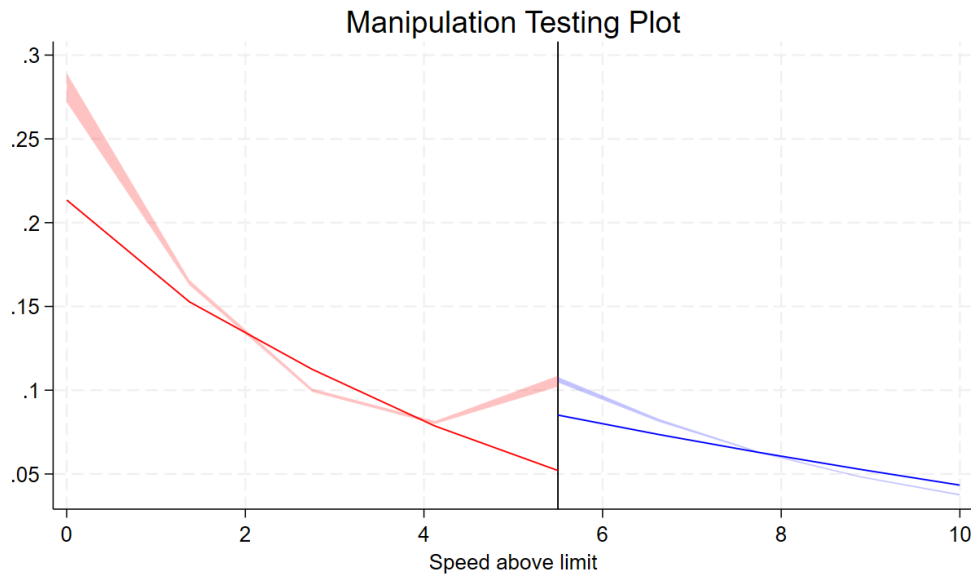
Another way to show that drivers are not selecting themselves into not receiving a ticket is to see what happens at the posted limit. One can see that there is bunching to the left of the posted limit, but to the right of the real limit.



Graph 5. Discontinuity around posted and real limit.

More formal evidence of this can be given by running the manipulation test developed by (Cattaneo et al., 2018). This is because of the discrete values of the assignment variable that does not allow for a McCrary test to be performed. The p-value for this test is 0.6461, so we can reject any manipulation. Result is the same for the matched sample, with a p-value for this test of 0.9251.

Graphs in the Appendix perform this test with different decision regarding the polynomial degree (Graph 4) or bandwidth (Graph 5). In both these cases the test is significant, but as the graphical evidence shows there is a discontinuity to the right of the cutoff. This is no manipulation since in principle drivers do not want to get a ticket, so it is unlikely to invalidate the exogeneity of the real limit.



Graph 6. Manipulation test based on discontinuity. Sample with polynomial of degree 2 and a bandwidth of 5 km/h below the real limit and 10 km/h above.

3.3 Smoothness on observables.

3.3.1 Full sample:

Another element I need to evaluate the validity of the fuzzy regression discontinuity design is to measure the smoothness of observable variables of the vehicles involved. I will do this first on the full sample, and then on the matched sample. For all estimations I use a polynomial of degree 2, and a bandwidth of 4.5 below and 10.5 above the cutoff (vehicles driving between 51 and 65 Km/h) with a triangular kernel. This leaves me with 136533 effective observations (76528 below and 60005 above the cutoff), of vehicles that only exceed the speed limit once during the sample period and did not receive a speeding ticket on the 90 days before doing so.

The first variable I explore is the hour of the day at which the ticket is received, a possible indications that drivers are different in the treatment and control groups. The same selection can be analyzed with day of the week as a continuous variable or a dummy if the ticket was given during the weekend. Since in Colombia, plates for motorcycles have a different configuration than those of other vehicles⁹, I can observe the type of vehicle involved. I use a dummy that takes the value of 1 if the vehicle involved is a motorcycle. Selection is a big concern when analyzing fines, and is the reason I estimate an RD in the first place. In any case, it is important to observe that treated and not treated do not differ in their usage of the road. I compare the number of times each vehicle passes under a Speed Camera from August, 2024

⁹ Motorcycle plates end in a letter, whether all other vehicle's plates end in a number.

to February 2025. As Table 7 shows, the only difference that one can observe is by vehicle type, and it is still small, less than 1%.

Table 7: Smoothnes of observables Full sample

	(1) Hour	(2) Day	(3) Weekend	(4) Motorcycle	(5) Observations
Bias-corrected	0.211 (0.214)	0.0923 (0.128)	-0.00149 (0.0292)	-0.0737** (0.0291)	1.466 (1.858)
<i>N</i>	136533	136533	136533	136533	136533
Mean	11.80	4.156	0.350	0.341	23.90

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 7. Smoothness on observables. Full sample. Hour is a continuous variable with the time in which the vehicle passed under the camera. Day is a continues variable that goes from 1 (Monday) to 7 (Sunday). Weekend is a dummy that takes the value of 1 for Saturday and Sunday, and 0 otherwise. Motorcycle is a takes the value of 1 for motorcycles, and 0 otherwise.

3.3.2 Matched sample:

In this case the total number of observations is 67,235 (39254 below and 27981 above the cutoff)¹⁰. Just like with the full sample, most of the variables are smooth at the cutoff. It is also the case that there are fewer motorcycles are the cutoff, but one should note the significant difference in percentages between the two samples. Whereas in the Full Sample 34% of vehicles are motorcycles, this percentage is only 7% in the matched sample. Note also that the Matched Sample has a higher number of observations, meaning that these vehicles are more frequent users of the road than the general sample. Both facts reinforce my argument about the Matched sample not being a representative part of the Full Sample.

Note that there is no significant difference in terms of price. This is central to my estimation later on.

¹⁰ As I explained later, the vast majority of those (62,288 or 92% of the matched sample) correspond to vehicles that are registered in Bogotá. The rest are registered in other municipalities but pay an exemption for the congestion restriction that exists in the city.

Table 8: Smoothnes of observables Matched sample

	(1) Hour	(2) Day	(3) Weekend	(4) Motorcycle	(5) Observations	(6) Price
Bias-corrected	-0.326 (0.248)	0.0615 (0.154)	-0.0164 (0.0351)	-0.0431** (0.0182)	1.833 (2.439)	-2.942 (3.817)
<i>N</i>	67235	67235	67235	67235	67235	67235
Mean	11.95	4.320	0.402	0.0702	31.13	56.86

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 8. Smoothness on observables. Full sample. Hour is a continuous variable with the time in which the vehicle passed under the camera. Day is a continues variable that goes from 1 (Monday) to 7 (Sunday). Weekend is a dummy that takes the value of 1 for Saturday and Sunday, and 0 otherwise. Motorcycle is a takes the value of 1 for motorcycles, and 0 otherwise.

4. General results.

In this section I show all the results. In section 4.1 I presents all results on reoffending – speeding. In Section 4.2 I presents results on Behavioral change – accidents. Section 4.3 is dedicated to the relative price analysis.

4.1 Reoffending – Speeding:

4.1.1. Full sample:

Table 9 presents the results for equation (1) with different dependent variables related to speeding for a period of six month after a vehicle is observed speeding under a Camera. Speeding relates to the fact that a vehicle was observed at a speed higher than 50 km/h at least once. Serious speeding (or Speeding>10) refers to the fact that a vehicle was observed at a speed higher than 60 km/h at least once.

This are the most important results of the paper. The results point at ticket having a causal effect of reducing all types of speeding behavior. This reduction is significant in magnitude. It is 64% for ever having a speeding infraction to 94% reduction for having a serious one.

Table 9: Effect of receiving a ticket

	(1) Speeding dummy	(2) Speeding continuous	(3) Speeding >10 dummy	(4) Speeding >10 continuous
Bias-corrected	-0.145*** (0.0255)	-0.237*** (0.0553)	-0.0538*** (0.0141)	-0.0663*** (0.0218)
Mean of not ticketed	0.226	0.370	0.0555	0.0716
N	136533	136533	136533	136533
N Left of cutoff	76528	76528	76528	76528
N Right of cutoff	60005	60005	60005	60005
BW left	5.040	5.040	5.040	5.040
BW right	10.60	10.60	10.60	10.60

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 9. Reoffending. Full sample. Column 1 (speeding) is a dummy that takes the value of 1 if the vehicle had at least one speeding, and 0 otherwise. Column 2 (Times speeding) is a continuous variable, the total of times a vehicle had at least one speeding. Column 3 (Speeding>10) is a dummy that takes the value of 1 if the vehicle had at least one serious speeding (above 60 km/h), and 0 otherwise. Column 4 (Times speeding>10) is a continuous variable, the total of times a vehicle had at least one serious speeding.

As I have said before, I am using a polynomial of degree 2 and a different bandwidth below and above the cutoff to have a comparable number of observations. Table 6 and 7 of the Appendix show that the results of Table 9 are robust to both decisions.

4.1.2 Full sample, by vehicle type:

A central question for policy makers is the effectiveness of speeding tickets and other measures for speed control in the face of the growing number of motorcycles in middle income countries. For simplicity I only make a dummy that says if a vehicle is a motorcycle (=1) or not (=0), and I will call not-motorcycles as cars. As it is shown in the Appendix, even if the discontinuity is much smaller for motorcycles, it is still present for both values of my dummy variable.

As Table 10 and 11 show, the general effect found on speeding is really only captured by cars. None of the coefficients for motorcycles are significant.

Vehicle type:

Table 10: Effect of receiving a ticket. Cars

	(1) Speeding dummy	(2) Speeding continuous	(3) Speeding >10 dummy	(4) Speeding >10 continuous
Bias-corrected	-0.207*** (0.0327)	-0.312*** (0.0730)	-0.0652*** (0.0176)	-0.0785*** (0.0273)
Mean of not ticketed	0.250	0.418	0.0571	0.0736
<i>N</i>	89037	89037	89037	89037
N Left of cutoff	51813	51813	51813	51813
N Right of cutoff	37224	37224	37224	37224
BW left	5.040	5.040	5.040	5.040
BW right	10.60	10.60	10.60	10.60

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 10. Reoffending. Full sample: Cars. Column 1 (speeding) is a dummy that takes the value of 1 if the vehicle had at least one speeding, and 0 otherwise. Column 2 (Times speeding) is a continuous variable, the total of times a vehicle had at least one speeding. Column 3 (Speeding>10) is a dummy that takes the value of 1 if the vehicle had at least one serious speeding (above 60 km/h), and 0 otherwise. Column 4 (Times speeding>10) is a continuous variable, the total of times a vehicle had at least one serious speeding.

Table 11: Effect of receiving a ticket. Motorcycles

	(1) Speeding dummy	(2) Speeding continuous	(3) Speeding >10 dummy	(4) Speeding >10 continuous
Bias-corrected	-0.0411 (0.0396)	-0.115 (0.0800)	-0.0324 (0.0238)	-0.0449 (0.0361)
Mean of not ticketed	0.177	0.274	0.0524	0.0678
<i>N</i>	47496	47496	47496	47496
N Left of cutoff	24715	24715	24715	24715
N Right of cutoff	22781	22781	22781	22781
BW left	5.040	5.040	5.040	5.040
BW right	10.60	10.60	10.60	10.60

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 11. Reoffending. Full sample: Motorcycles. Column 1 (speeding) is a dummy that takes the value of 1 if the vehicle had at least one speeding, and 0 otherwise. Column 2 (Times

speeding) is a continuous variable, the total of times a vehicle had at least one speeding. Column 3 (Speeding>10) is a dummy that takes the value of 1 if the vehicle had at least one serious speeding (above 60 km/h), and 0 otherwise. Column 4 (Times speeding>10) is a continuous variable, the total of times a vehicle had at least one serious speeding.

4.1.3. Matched sample:

I can also do this analysis in the matched sample. As Table 12 shows the results are essentially the same, a general effect of reduction in all types of speeding. Note also that the result is slightly less robust than in the general sample.

Matched sample:

Table 12: Effect of receiving a ticket

	(1) Speeding dummy	(2) Speeding continuous	(3) Speeding >10 dummy	(4) Speeding >10 continuous
Bias-corrected	-0.138*** (0.0311)	-0.180*** (0.0628)	-0.0399** (0.0165)	-0.0456** (0.0229)
Mean of not ticketed	0.258	0.405	0.0565	0.0687
N	67235	67235	67235	67235
N Left of cutoff	39254	39254	39254	39254
N Right of cutoff	27981	27981	27981	27981
BW left	5.040	5.040	5.040	5.040
BW right	10.60	10.60	10.60	10.60

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 12. Reoffending. Matched sample. Column 1 (speeding) is a dummy that takes the value of 1 if the vehicle had at least one speeding, and 0 otherwise. Column 2 (Times speeding) is a continuous variable, the total of times a vehicle had at least one speeding. Column 3 (Speeding>10) is a dummy that takes the value of 1 if the vehicle had at least one serious speeding (above 60 km/h), and 0 otherwise. Column 4 (Times speeding>10) is a continuous variable, the total of times a vehicle had at least one serious speeding.

Since my matched sample is almost exclusively comprised of not-motorcycles, there is no point in doing this analysis by vehicle type.

4.2 Behavioral change – accidents:

4.2.1 Full sample:

Table 13 presents the results of equation (1) when the dependent variable measures accidents. Neither a dummy for having an accident (column 1), nor the continuous

variable for total of accidents (column 2), nor the dummy for accident with at least one fatality (column 3) are significant. Nor is it useful to separate the sample into cars and motorcycles.

Table 13: Effect of receiving a ticket

	(1) Any accident dummy	(2) Total Accidents continuous	(3) Deadly accidents dummy	(4) Cars dummy	(5) Motorcycles dummy
Bias-corrected	-0.00624 (0.00476)	-0.00653 (0.00486)	-0.00118 (0.000843)	-0.00450 (0.00585)	-0.00962 (0.00820)
Mean of not ticketed	0.00590	0.00599	0.000211	0.00577	0.00614
<i>N</i>	136533	136533	136533	89037	47496
<i>N</i> Left of cutoff	76528	76528	76528	51813	24715
<i>N</i> Right of cutoff	60005	60005	60005	37224	22781
BW left	5.04	5.04	5.04	5.04	5.04
BW right	10.60	10.60	10.60	10.60	10.60

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 13. Behavioral change. Full sample. Column 1 (Any accident) is a dummy that takes the value of 1 if the vehicle had at least one accident in the following months, and 0 otherwise. Column 2 (Total Accidents) is a continuous variable; the total of times a vehicle had an accident. Column 3 (Deadly Accident) is a dummy that takes the value of 1 if the vehicle had at least one accident with a fatality in the following months, and 0 otherwise. Column 4 (Cars) is the same as Column 1 restricting the sample to cars. Column 5 (Motorcycles) does the same for motorcycles.

A possible objection to these results could be the fact that the estimation is underpowered. I should not that the same number of observations produced a very credible result for speeding. I should also point out that there are 724 accidents in my estimation sample. These are around 6% of the total 11922 total accidents that occurred in the period of study. One could extend the sample some additional months to obtain a bigger number of accidents. When widening the time period, even if the total of accidents in the sample increases to 892 (one additional month) or 928 (two months), the result (not reported) does not change.

4.2.2 Matched sample:

The result holds if one looks at the Matched sample. As can be seen in Table 14, results are essentially the same; although point estimates are negative, they are far from significant.

Table 14: Effect of receiving a ticket

	(1) Any accident dummy	(2) Total accidents continous	(3) Deadly accidents dummy
Bias-corrected	-0.000634 (0.00574)	-0.000435 (0.00576)	-0.000441 (0.000958)
Mean of not ticketed	0.00603	0.00611	0.000165
<i>N</i>	67235	67235	67235
N Left of cutoff	39254	39254	39254
N Right of cutoff	27981	27981	27981
BW left	5	5	5
BW right	10.60	10.60	10.60

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 14. Behavioral change. Matched sample. Column 1 (Any accident) is a dummy that takes the value of 1 if the vehicle had at least one accident in the following months, and 0 otherwise. Column 2 (Total Accidents) is a continuous variable, the total of times a vehicle had an accident. Column 3 (Deadly Accident) is a dummy that takes the value of 1 if the vehicle had at least one accident with a fatality in the following months, and 0 otherwise.

A natural question for the lack of results on accidents is the fact that my estimation could be underpowered. The following simulations try to answer that concern, but there is in fact a way to find an effect on accidents: divide the sample into frequent and infrequent drivers. I will go deeper on this on section 7, but for now one can say that if my sample size is sufficient for finding a deterrent effect on speeding is underpowered for finding an effect on accidents, that is a finding in itself.

Using the *rdpow* command in Stata, I can calculate that my sample size has a power of 0.965 for the coefficient found for Speeding (column 1 table 9). In other words, it could even detect an effect 20% smaller. On the contrary, my sample size is underpowered for finding an effect on accidents (column 1, table 6). I only have a power of 0.142 for the coefficient found. In fact, I would need the coefficient to be more than three times larger in order to have more than the conventional 0.8 of power.

5. Relative price.

One of the central questions of this paper is if fixed monetary fines are equally effective. Many countries have fines that are relative to income, mainly out of justice considerations. I am thinking mainly about effectiveness. Table 15 reproduces equation (1) with the dependent variable of a dummy of speeding (column 1 of table

8). Remember that the quintile corresponds to the commercial price of the vehicle as proxy for the owner's income. One can see that the effect is concentrated on the higher quintiles of the distribution. Both magnitude and significance are lower for the lower quintiles.

Table 15: Effect of receiving a ticket by quintile

	(1) Any speeding Quintile 1	(2) Any speeding Quintile 2	(3) Any speeding Quintile 3	(4) Any speeding Quintile 4	(5) Any speeding Quintile 5
Bias-corrected	-0.108 (0.0984)	-0.0884 (0.0663)	-0.157** (0.0751)	-0.123** (0.0575)	-0.178*** (0.0649)
Mean of not ticketed	0.170	0.240	0.244	0.263	0.278
<i>N</i>	7065	11848	12824	17225	18273
<i>N</i> Left of cutoff	3901	7055	7617	10142	10539
<i>N</i> Right of cutoff	3164	4793	5207	7083	7734
BW left	5.04	5.04	5.04	5.04	5.04
BW right	10.60	10.60	10.60	10.60	10.60

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 15. Reoffending by quintile of vehicle price. Matched sample. Dependent variable is a dummy that takes the value of 1 if the vehicle had at least one speeding, and 0 otherwise.

Table 16 does the same for the dummy of ever being in an accident. Only one significant coefficients is found, and in this case in the undesired direction. It seems that for quintile 2 of the income distribution, speeding tickets do not reduce speed but in fact increase the chances of being in an accident. One should not overinterpret a coefficient, but it is not something to be ignored coupled with the fact that Quintile 2 was also the one where speeding fines were less deterrent of speeding.

Table 16: Effect of receiving a ticket by quintile

	(1) Any accident Quintile 1	(2) Any accident Quintile 2	(3) Any accident Quintile 3	(4) Any accident Quintile 4	(5) Any accident Quintile 5
Bias-corrected	-0.0129 (0.0224)	0.0339*** (0.00490)	-0.00504 (0.0126)	-0.00613 (0.0110)	-0.0141 (0.0138)
Mean of not ticketed	0.00608	0.00282	0.00484	0.00578	0.00857
<i>N</i>	7065	11848	12824	17225	18273
<i>N</i> Left of cutoff	3921	6754	7121	9295	9292
<i>N</i> right of cutoff	3157	4601	4872	6455	6820
BW left	5	5	6	5	6
BW right	10.60	10.60	10.60	10.60	10.60

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 16. Behavioral change by quintile of vehicle price. Matched sample. Dependent variable is a dummy that takes the value of 1 if the vehicle had at least one serious speeding (above 60 km/h), and 0 otherwise.

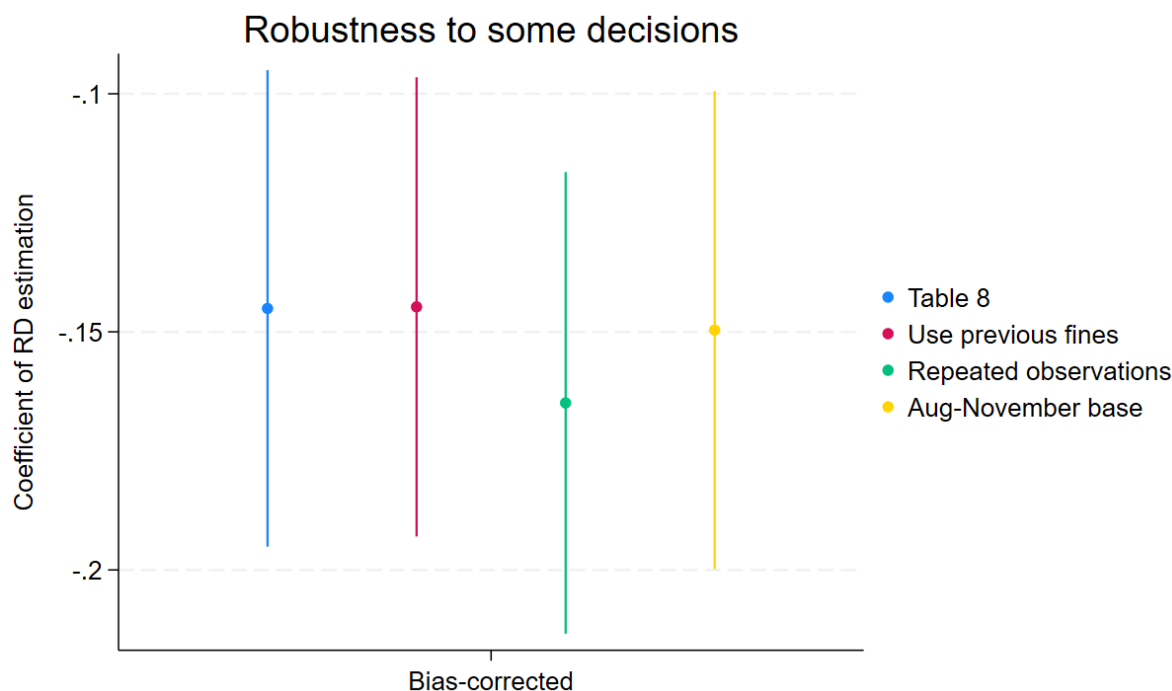
6. Robustness checks.

In this section I introduce some further evidence that my results are credible. First, I present alternative estimations to show consistency. Second, I use an alternative method of estimating my results: an IV-judge type of estimation using data on the police agent in charge of verifying each speeding ticket.

6.1 Robustness to some decisions:

Graph 17 shows a comparison between the coefficient found on column 1 of Table 8 and the coefficient found with different decisions. In particular, whether to include vehicles that had a previous fine, to use vehicles that pass more than once under a Camera and to have a bigger time for selecting my sample.

The coefficient is very stable across all types of estimations, and bigger for the case in which vehicles that passed more than once are included. This is discussed in bigger detail in section 2 of the Appendix, where I also build a bigger database.

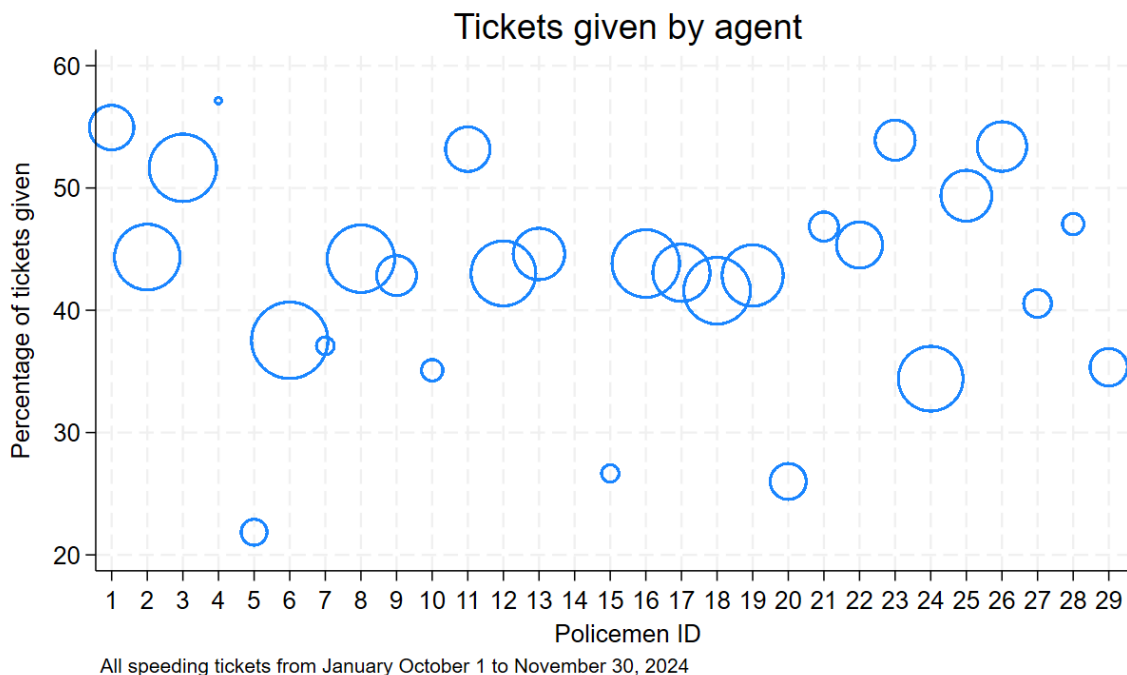


Graph 17. “Use previous fines” uses the same estimation, this time including vehicles that have had a ticket in the previous six months. “Repeated observations” shows the coefficient using Vehicles that passed more than once, as it is further discussed in section 2 of the Appendix. “Aug-November” base makes the database bigger, by including al vehicles from August 16 to November 15, 2024.

6.2 An IV-judge using police officer manual validation:

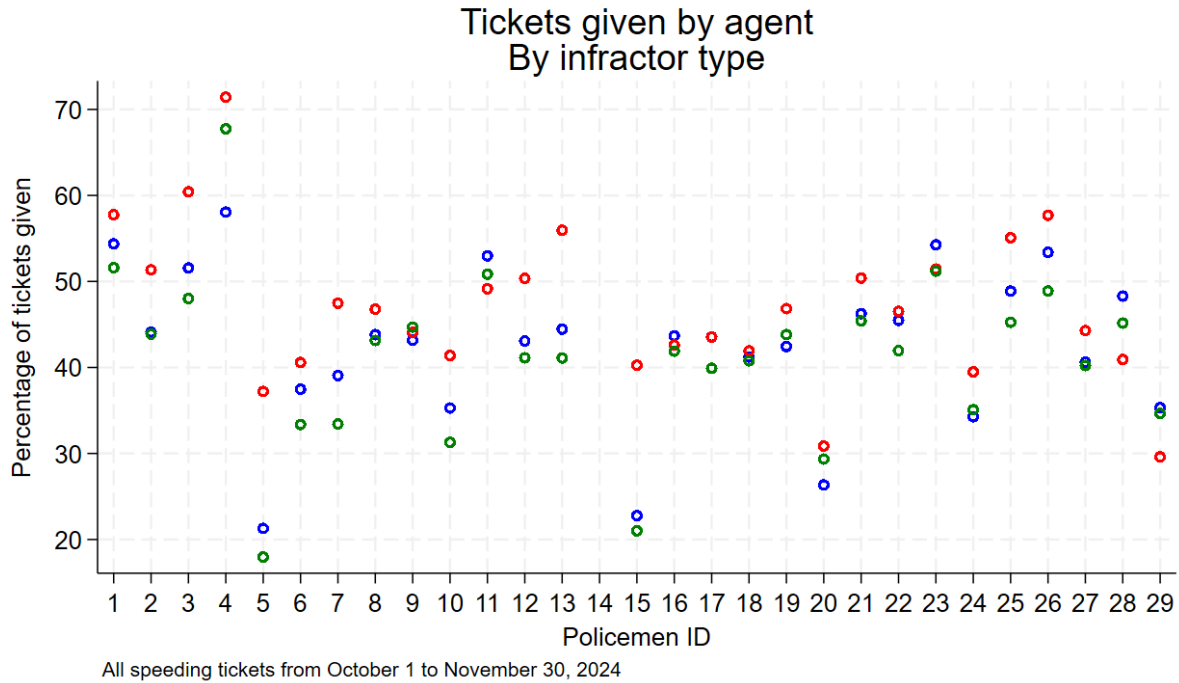
According to Colombian law all speeding tickets must be subject to human verification of their validity. This has to do mainly with the fact that the technology for taking pictures of vehicles while speeding is not perfect, and some pictures are blurry. This task is done by trained police officers that must evaluate whether a specific picture should transform into a ticket. The pictures assigned to each policemen are random, and they have no way to refuse their verification.

I have data for the identification of all pictures of vehicles that were circulating above 56 km/h during the months of October and November. I also can see which agent was in charge of reviewing each picture. As Graph 18 shows, there is considerable heterogeneity in the percentage of pictures that each agent transforms into a ticket.



Graph 18. Policeman ID is a number to identify each police officer. The size of the circle is proportional to the number of pictures reviewed by each agent.

Graph 19 indicates that there might be some randomness associated to the Policemen in charge. For this to transform into a credible instrumental variable one would want to verify that these different rates are not related to any observable characteristic of the ticket, or the vehicle involved. Graph 20 shows that the strictness is stable for all Policemen across at least two meaningful variables, whether a vehicle is a motorcycle and those vehicles that drive faster than 60 km/h.



Graph 19. Policeman ID is a number to identify each police officer. Blue dots correspond to all pictures, red dots to pictures of motorcycles and green to people under serious speeding (40% of this subsample).

Both previous graph give evidence to use the random assignment of the Policemen in charge of reviewing each picture as an instrument of the ticket. I will then capture the part of receiving a ticket that is due to the randomness of the assigned policemen. I also must emphasize that this is a different sample, consisting mostly of drivers that decide to exceed speeding limits seriously. I should also note that in this case I am not finding a local effect, but the general effect on those that decide to exceed the limit by a significant amount of kilometers.

Table 20 shows the result of an estimation with the same dependent variables as Table 8. As one can clearly see, the significance and sign of all the estimation is similar, but magnitudes are considerably smaller. While the effect of a speeding ticket was a 64% reduction in the change of reoffending, in this case the effect is only 25%. This surely is caused by these drivers deciding to disobey speeding limits by a bigger amount than the other sample.

Table 20: Effect of agent reviewing ticket

	(1) Speeding dummy	(2) Speeding continuous	(3) Speeding >10 dummy	(4) Speeding >10 continuous
Ticket	-0.0485*** (0.0170)	-0.105*** (0.0331)	-0.0194** (0.00894)	-0.0246* (0.0126)
Mean of not ticketed	0.196	0.310	0.0470	0.0604
<i>N</i>	132668	132668	132668	132668
First stage Wald-F	82.56	82.56	82.56	82.56

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 20. Reoffending. Full sample. Column 1 (speeding) is a dummy that takes the value of 1 if the vehicle had at least one speeding, and 0 otherwise. Column 2 (Times speeding) is a continuous variable, the total of times a vehicle had at least one speeding. Column 3 (Speeding>10) is a dummy that takes the value of 1 if the vehicle had at least one serious speeding (above 60 km/h), and 0 otherwise. Column 4 (Times speeding>10) is a continuous variable, the total of times a vehicle had at least one serious speeding. This table is comparable to Table 9.

The results on Table 20 are in line with the main finding of this paper. One could also use this sample to verify the second finding: effects being explained only by cars, not motorcycles. Table 21 and Table 22 present the same estimation. Note that in this case the effect is more precisely estimated for cars than motorcycles, but there is some for the latter. Once again, this does not cast doubt on previous findings but shows that serious speeders are more similar between them, independent of the type of vehicle they drive.

Table 21: Effect of agent reviewing ticket. Cars

	(1) Speeding dummy	(2) Speeding continuous	(3) Speeding >10 dummy	(4) Speeding >10 continuous
Ticket	-0.0487** (0.0196)	-0.117*** (0.0384)	-0.0159 (0.00994)	-0.0236* (0.0138)
Mean of not ticketed	0.209	0.332	0.0451	0.0572
<i>N</i>	94536	94536	94536	94536
First stage Wald-F	65.40	65.40	65.40	65.40

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 21. Reoffending. Full sample. Column 1 (speeding) is a dummy that takes the value of 1 if the vehicle had at least one speeding, and 0 otherwise. Column 2 (Times speeding) is a continuous variable, the total of times a vehicle had at least one speeding. Column 3 (Speeding>10) is a dummy that takes the value of 1 if the vehicle had at least one serious speeding (above 60 km/h), and 0 otherwise. Column 4 (Times speeding>10) is a continuous variable, the total of times a vehicle had at least one serious speeding. This table is comparable to Table 10.

Table 22: Effect of agent reviewing ticket. Motorcycles

	(1) Speeding dummy	(2) Speeding continuous	(3) Speeding >10 dummy	(4) Speeding >10 continuous
Ticket	-0.0453* (0.0269)	-0.0434 (0.0491)	-0.0273* (0.0159)	-0.0280 (0.0229)
Mean of not ticketed	0.163	0.252	0.0519	0.0691
<i>N</i>	38132	38132	38132	38132
First stage Wald-F	28.93	28.93	28.93	28.93

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 22. Reoffending. Full sample. Column 1 (speeding) is a dummy that takes the value of 1 if the vehicle had at least one speeding, and 0 otherwise. Column 2 (Times speeding) is a continuous variable, the total of times a vehicle had at least one speeding. Column 3 (Speeding>10) is a dummy that takes the value of 1 if the vehicle had at least one serious speeding (above 60 km/h), and 0 otherwise. Column 4 (Times speeding>10) is a continuous variable, the total of times a vehicle had at least one serious speeding. This table is comparable to Table 11.

One could use this estimation strategy to verify the other two findings. As Table 23 shows there is also a null effect on all dependent variables related to the chance of being involved in an accident. In this case the coefficients are positive, not like those found in my RD estimation that were negative but not significant.

Table 23: Effect of agent reviewing ticket

	(1)	(2)	(3)	(4)	(5)
	Any accident dummy	Total Accidents continuous	Deadly accidents dummy	Cars dummy	Motorcycles dummy
Ticket	0.000223 (0.09)	0.000417 (0.17)	0.000590 (1.24)	0.000807 (0.31)	-0.00439 (-0.92)
Mean of not ticketed	0.00377	0.00380	0.000145	0.00362	0.00419
N	132668	132668	132668	94536	38132
First stage Wald-F	82.56	82.56	82.56	65.40	28.93

t statistics in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 23. Behavioral change. Full sample. Column 1 (Any accident) is a dummy that takes the value of 1 if the vehicle had at least one accident in the following months, and 0 otherwise. **Column 2 (Total Accidents)** is a continuous variable; the total of times a vehicle had an accident. **Column 3 (Deadly Accident)** is a dummy that takes the value of 1 if the vehicle had at least one accident with a fatality in the following months, and 0 otherwise. **Column 4 (Cars)** is the same as Column 1 restricting the sample to cars. **Column 5 (Motorcycles)** does the same for motorcycles. This table is comparable to Table 14.

The final finding one could verify has to do with the heterogeneous effect by vehicle price. As Table 24 shows the effect of a ticket on speeding behavior is in this case is also concentrated on the higher quintiles. Even more, like in previous cases for quintile 2 there is a slightly positive coefficient.

Table 24: Effect of agent reviewing ticket. By quintile

	(1)	(2)	(3)	(4)	(5)	(6)
	Matched Sample	Quintile 1	Quintile 2	Quintile 3	Quintile 4	Quintile 5
Ticket	-0.0828*** (0.0240)	-0.0936 (0.0742)	0.102* (0.0530)	-0.0962** (0.0446)	-0.136*** (0.0473)	-0.129*** (0.0464)
Mean of not ticketed	0.234	0.150	0.210	0.227	0.246	0.266
N	48230	3562	8772	10223	12150	13523
First stage Wald-F	46.28	3.551	8.688	13.11	12.64	13.37

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 24. Reoffending by price quintile. Matched sample. Dependent variable is a dummy that takes the value of 1 **Column 1 (speeding)** is a dummy that takes the value of 1 if the vehicle had at least one speeding, and 0 otherwise

In summary, although the sample for the previous sections is different from this one, nor are the estimation strategies capturing the same causal effect, one can safely

say that these results give more credibility to the causal effects found. All evidence points in the same direction.

7. Wake-up call or information? Frequency of driving as mechanism.

Speeding tickets are meant to prevent the most serious consequences of traffic accidents. In fact, speeding Cameras in Bogotá were once called “*Cameras Salvavidas*” (*Lifesaving cameras*). As we have seen, in practice Cameras seem very effective at reducing speeding itself but show little effect on reducing accidents.

These effects can be mediated by different types of drivers. A natural distinction is the frequency by which each driver uses the road. Frequent drivers both know speed limits, and Camera location, better than infrequent drivers. Thus one could imagine that the effect of a fine will be lower for them on their reoffending probabilities. Fines are less of a way of acquiring new information for them.

It is hard to imagine, in principle, how frequency of driving mediates on the effect of a fine on driving behavior and thus probability of being in an accident. A natural intuition is that frequency of driving gives experience and experience gives drivers better control over their abilities. Thus the State will be less eager to tell experienced drivers to not speed on the road. On the contrary, this message will be more important for less frequent drivers that might be more conscious of their limited abilities.

Table 25 and 26 present my main estimation by frequency of road usage. Even if for my RD I only use vehicles that are circulating above the posted speed limit, I have the full data for all vehicles circulating under a Camera for six months (August 16, 2024 to February 15, 2025). With such data I can see the total number of times a vehicle circulates. I will define vehicles as high (low) frequency if the total number of observations is higher (lower) than the median. To make the cut less sharp I will not use of observations located between percentile 45 and 55, in order to have sufficiently different groups.

Table 25 presents results for Speeding. Note that there are effects for both types of drivers and for minor and serious speeding. Consistent with a notion that a speeding ticket gives much more information to a less frequent driver, the magnitude of the effect is bigger for low frequency drivers. It is worth noting the results for all groups on serious speeding, since it points to fines giving information even to frequent drivers. This coincides with the fact that the rates at which both types of drivers receive a ticket is similar, 18.7% for low frequency drivers and 15.2% for high frequency drivers, although naturally higher for low frequency drivers. But the difference in driving frequency is of orders of magnitude.

Table 25: Effect of a ticket by frequency of driving

	(1) Speeding All drivers	(2) Speeding Low frequency	(3) Speeding High frequency	(4) Speeding >10 All drivers	(5) Speeding >10 Low frequency	(6) Speeding >10 High frequency
Bias-corrected	-0.145*** (0.0255)	-0.0683** (0.0275)	-0.212*** (0.0415)	-0.0538*** (0.0141)	-0.0388** (0.0156)	-0.0727*** (0.0242)
Mean of not ticketed	0.226	0.0961	0.353	0.0555	0.0270	0.0837
Frequency (mean)	24.62	5.072	43.34	24.62	5.072	43.34
<i>N</i>	136533	64075	62618	136533	64075	62618
N Left of cutoff	76528	33745	37223	76528	33745	37223
N Right of cutoff	60005	30330	25395	60005	30330	25395
BW left	5.040	5.040	5.040	5.040	5.040	5.040
BW right	10.60	10.60	10.60	10.60	10.60	10.60

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 25. Reoffending. Full sample. Column 1 to 3 (speeding) are a dummy that takes the value of 1 if the vehicle had at least one speeding, and 0 otherwise. Column 4 to 6 (Speeding>10) are a dummy that takes the value of 1 if the vehicle had at least one serious speeding (above 60 km/h), and 0 otherwise.

Table 26 presents results for Accidents. As I said when I discussed the general finding, the only group for which one finds a deterrent effect of tickets on accidents is low-frequency drivers. In fact the coefficient is even positive, although not significant, for high frequency. A natural way of reading this number is that high frequency drivers react to a speeding fine by overspeeding on all locations where they know there are no speed Cameras. Another possible interpretation is that a speeding fine breeds frustration and thus higher probability of an accident.

Table 26: Effect of a ticket by frequency of driving

	(1) Any accident All drivers	(2) Any accident Low- frequency drivers	(3) Any accident High frequency drivers
Bias-corrected	-0.00624 (0.00476)	-0.0130** (0.00618)	0.00217 (0.00761)
Mean of not ticketed	0.00583	0.00369	0.00815
Frequency (mean)	24.62	5.072	43.34
<i>N</i>	136533	64075	62618
N Left of cutoff	76528	33745	37223
N Right of cutoff	60005	30330	25395
BW left	5	5	5
BW right	10.60	10.60	10.60

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 26. Behavioral change by frequency. Full sample. Dependent variable is a dummy that takes the value of 1 if the vehicle had at least one serious speeding (above 60 km/h), and 0 otherwise.

I should also note that some estimations (not reported) show that this no-deterrence on the high frequency drivers is not caused by the most frequent drivers. Excluding only drivers above the 90th or 80th percentile does not make the coefficient significant at the 10% level.

8. Discussion and possible policy recommendations.

This paper can be summarized in one sentence: speeding tickets causally reduce speeding but seem to have a much smaller (probably null) effect in reducing accidents. As I argued, technically I have not enough statistical power to say that there is no effect on accidents, but I can confidently say that the effect is negative and statistically and practically significant for speeding.

The reduction of speeding is concentrated both in automobiles and lesser speeding infractions (this I will argue comes specifically from the judge-IV). This main finding is true both for the Full sample and for the Matched sample of vehicles that are registered in Bogotá. The effects on speeding are also concentrated on the higher quintiles of the distribution, and no effect of relative price is found for accidents. The only deterrent effect of speeding tickets on accidents is for those vehicles that use the roads less frequently.

Gary Becker wrote that “Fines are preferable to imprisonment and other types of punishment because they are more efficient” (Becker, 1992). In his analysis “the social cost of fines is about zero”, and the optimal amount of a fine “depend[s] only on the marginal harm and cost and not at all on the economic positions of offenders”(Becker, 1974). To be fair, Becker was surely also thinking of more serious crimes than a speed of 56 km/h, but those strong statements are also a clear representation of his thinking.

This paper shows that fines can be socially inefficient in a very particular way. Speeding fines are just one example of many where the State punishes a proxy behavior (speeding) because it cannot punish the most undesired outcome (a car accident). I think my evidence is strong in the direction that fines are not serving their main purpose. A fine that does not deter is not socially efficient. On top of that, it seems clear from Becker’s thinking that fines should not have a different deterrent effect depending on the income of the individual. This is also not the case in the context I study, and is an additional source of inefficiency since resources are only scarcely transferred to the State.

The second inefficiency might be particularly relevant for thinking about enforcement on developing countries. Becker’s logic might need to be complicated in contexts where the State can impose a fine but it can do little to enforce it. As I explained before, in the context of Bogotá the only real effect of not paying a speeding fine is

the exclusion from the formal market, both in the market for used vehicles and the formal labor market due to the risk of getting a wage garnishment for unpaid fines. Fixed monetary fines might turn into an additional barrier to formality for the poorest members of society.

All this points to a clear policy recommendation: relative fines. Seeding tickets might have a better effect if they were adjusted, to a lower value, for the less wealthy drivers. If one adds the known fact that higher quintiles have higher payment rates, one can conclude that the State will prevent more speeding and even collect more resources if fines are lower for the poorer. At the same time this might reduce the barrier to formality that existing fines can be.

I would argue that the main effect for public policy of the findings is that the presumed causal relation between speed and accidents is less important than assumed. If speeding tickets have a significant and negative effect on speeding behavior but they do not have any significant effect on accidents (even in one case an undesired effect), then speed is not a significant cause for accidents. This has a clear policy application: other measures are needed in order to prevent road accidents from becoming an even bigger public health challenge.

The fact that the desired effects of the policy are concentrated on automobiles, and absent in motorcycles also has important policy implications. This might mean that speeding tickets are not the right tool for the changing face of the public health challenge of road fatalities that some cities are facing, since motorcycle drivers are a growing percentage of road fatalities both as victims and perpetrators¹¹. Other interventions are needed if a city like Bogotá (and other in middle income countries) want to face the public health challenge of motorcycles (in all its dimensions: (Martínez-González et al., 2022)). Both non-automated control and behavioral interventions are urgently needed.

I don't think that an implication of the evidence shown is that speed cameras don't work. What this paper has shown is that they do not work through a very specific channel: the behavioral change of marginal speeders just above an unknown real fine threshold. Most evaluations seem to show that in the aggregate cameras work, and this might be easily explained by their effect on regular speed-limit obeying drivers both internationally (Ang et al., 2020) and specifically in Bogotá (Ferro, 2024). Newer policies might be needed for the selected population that knowing the speed limit and how strongly it is enforced, decides to speed. For the high majority of drivers knowing that there is police enforcement of speeding laws is sufficient to positively influence their behavior.

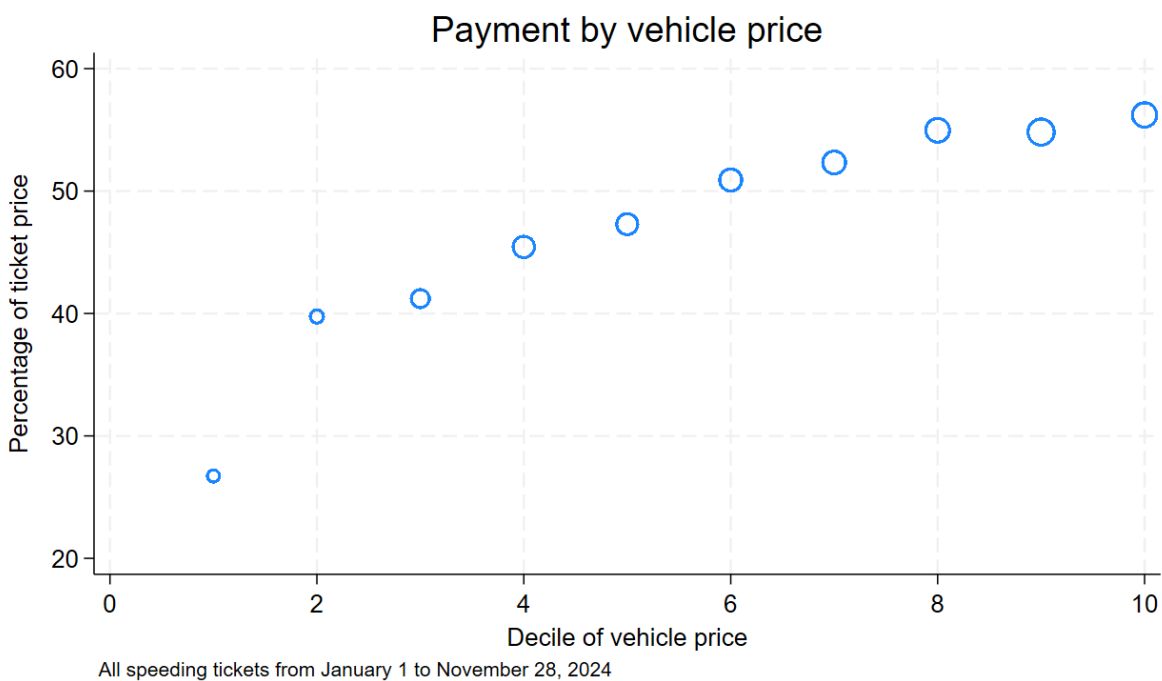
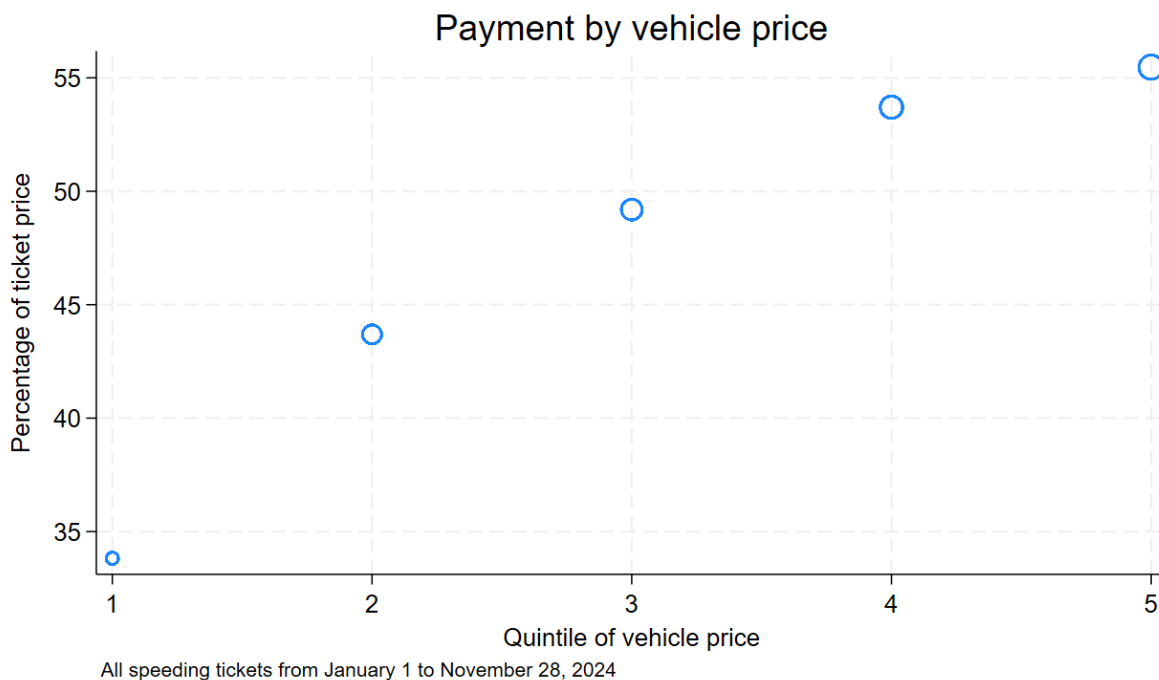
¹¹ In Bogotá, motorcycle users were present in 45.5% of accidents with a death in 2015. This number has steadily grown to 68.5% in 2024.

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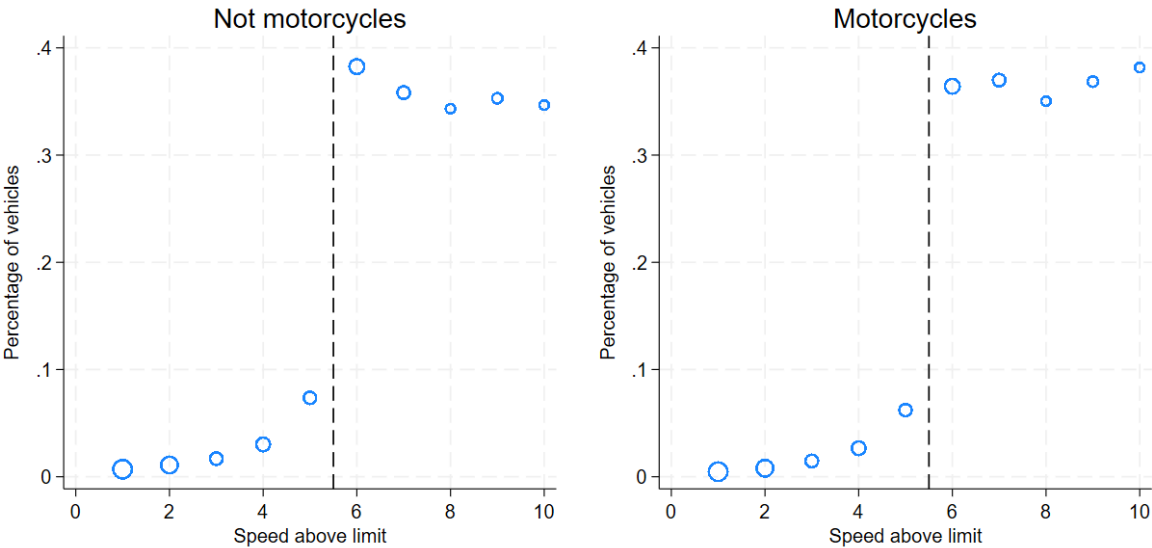
Appendix:



repetidos	Freq.	Percent	Cum.
1	158,703	87.44	87.44
2	17,731	9.77	97.21
3	3,460	1.91	99.12
4	934	0.51	99.64
5	371	0.20	99.84
6	142	0.08	99.92
7	52	0.03	99.95
8	39	0.02	99.97
9	21	0.01	99.98
10	10	0.01	99.99
11	13	0.01	99.99
12	3	0.00	99.99
13	6	0.00	100.00
15	1	0.00	100.00
17	1	0.00	100.00
18	1	0.00	100.00
26	1	0.00	100.00
Total	181,489	100.00	

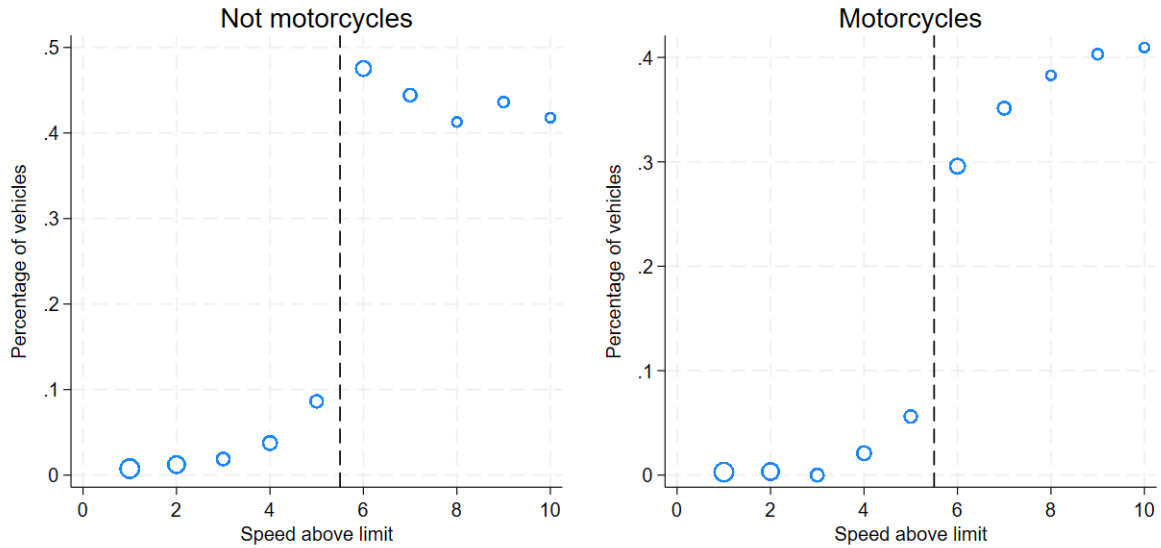
Table1. Times each plate passed exceeding the limit under the Cameras.

Percentage of vehicles that recieve a speeding ticket
by vehicle type
Full sample

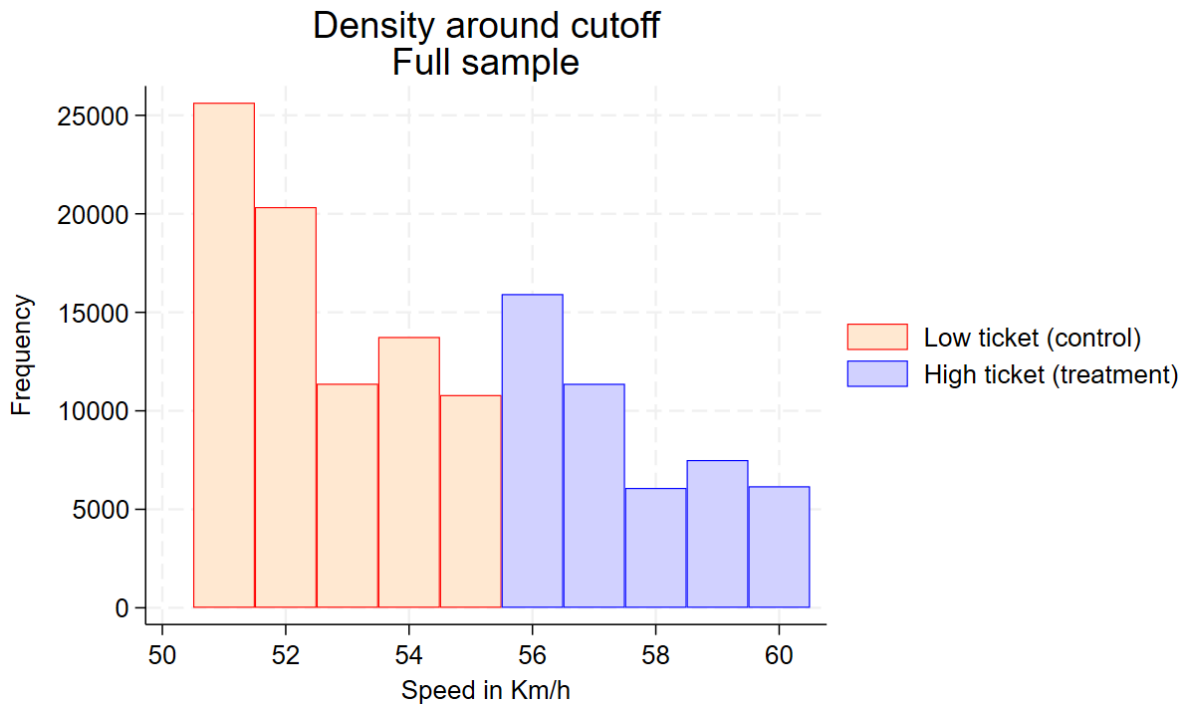


Graph 2.

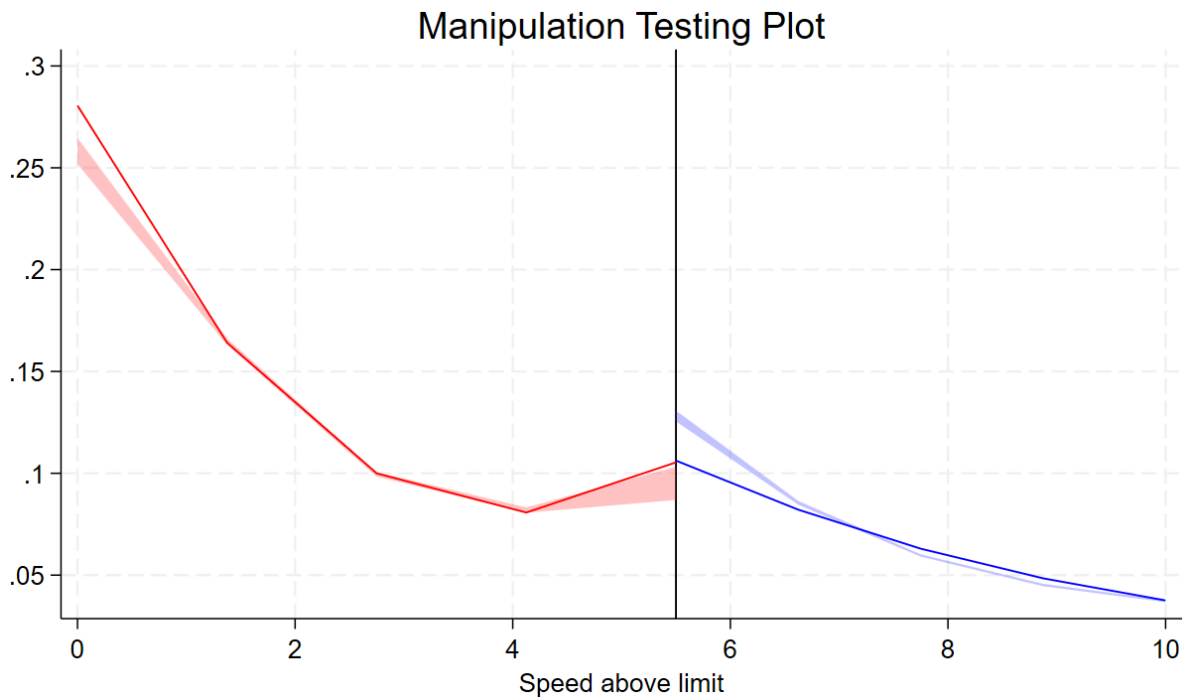
Percentage of vehicles that receive a speeding ticket by vehicle type Matched sample



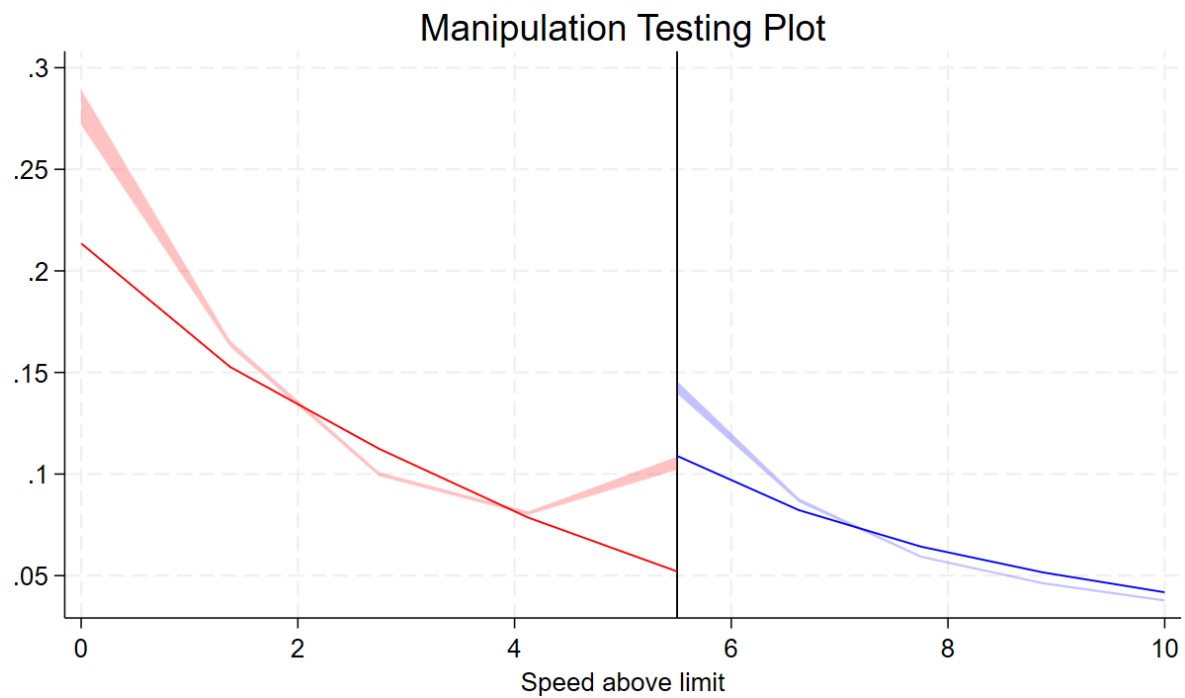
Graph 3.



Graph 4.



Graph 4. Manipulation test based on discontinuity. Sample with polynomial of degree 3 and a bandwidth of 5 km/h below the real limit and 10 km/h above.



Graph 5. Manipulation test based on discontinuity. Sample with polynomial of degree 2 and a bandwidth of 5 km/h below the real limit and 5 km/h above.

Table 6: Robustness to polynomial and bandwidth. Full sample

	(1) Speeding Table 9 (1)	(2) Polynomial of degree 3	(3) Bandwith bwselect	(4) Speeding >10 Table 9 (3)	(5) Polynomial of degree 3	(6) Bandwith bwselect
Bias-corrected	-0.145*** (0.0255)	-0.180*** (0.0396)	-0.0959*** (0.0284)	-0.0538*** (0.0142)	-0.0733*** (0.0219)	-0.0454*** (0.0157)
Mean of not ticketed	0.225	0.225	0.225	0.0567	0.0567	0.0567
<i>N</i>	136533	136533	136533	136533	136533	136533
N Left of cutoff	76528	76528	76528	76528	76528	76528
N Right of cutoff	60005	60005	43826	60005	60005	43826
BW left	5	5	5.040	5	5	5.040
BW right	10.60	10.60	5.040	10.60	10.60	5.040

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 6 Appendix. Reoffending. For columns 1 to 3, the dependent variable (speeding) is a dummy that takes the value of 1 if the vehicle had at least one speeding, and 0 otherwise. For columns 4 to 6 the dependent variable is a dummy that takes the value of 1 if the vehicle had at least one serious speeding (above 60 km/h), and 0 otherwise. Name of the column presents the variation with respect to the main estimation.

Table 7: Robustness to polynomial and bandwidth. Matched sample

	(1) Speeding Table 9 (1)	(2) Polynomial of degree 3	(3) Bandwith bwselect	(4) Speeding >10 Table 9 (3)	(5) Polynomial of degree 3	(6) Bandwith bwselect
Bias-corrected	-0.138*** (0.0311)	-0.207*** (0.0483)	-0.0783** (0.0345)	-0.0400** (0.0165)	-0.0656*** (0.0254)	-0.0445** (0.0183)
Mean of not ticketed	0.257	0.257	0.257	0.0572	0.0572	0.0572
<i>N</i>	67235	67235	67235	67235	67235	67235
N left of cutoff	39254	39254	39254	39254	39254	39254
N right of cutoff	27981	27981	20749	27981	27981	20749
BW left	5	5	5.040	5	5	5.040
BW right	10.60	10.60	5.040	10.60	10.60	5.040

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 7 Appendix. Reoffending. For columns 1 to 3, the dependent variable (speeding) is a dummy that takes the value of 1 if the vehicle had at least one speeding, and 0 otherwise. For columns 4 to 6 the dependent variable is a dummy that takes the value of 1 if the vehicle had at least one serious speeding (above 60 km/h), and 0 otherwise. The name of the column presents the variation with respect to the main estimation.

2. Vehicles that passed more than once under a speeding Camera.

In my main estimation I use only vehicles in my sample that pass under a Camera once. This is because observed speed compared to the real limit is the variable that randomly assigns vehicles to treatment, thus if a vehicle passes once below the real limit and once above the real limit is not clear to which condition of treatment it should be assigned. In the following I use two different strategies to include vehicles that pass more than once under a Camera. All of the results of the paper stand in both strategies.

A natural alternative, arguably the most common on the RD literature, is to use the first time a vehicle passes through the Cameras as the assignation to treatment. In this context it is not the best strategy since there is a time lapse of at least three and maybe even ten days between the time a vehicle commits the speeding infraction and receives the ticket. In Table 8 the same result of Table 9 is presented with this full sample. Note that there is an increase of 12% in observations (from 136553 to 152864), and those observations tend to commit higher number of speeding infractions. The coefficients are still significant in the extensive margin, but less so on the intensive one.

Table 8: Using first observation to assign to treatment

	(1) Speeding dummy	(2) Speeding continuous	(3) Speeding >10 dummy	(4) Speeding >10 continuous
Bias-corrected	-0.158*** (0.0266)	-0.149* (0.0761)	-0.0314** (0.0158)	-0.0292 (0.0297)
Mean of not ticketed	0.299	0.570	0.0802	0.113
<i>N</i>	152864	152864	152864	152864
<i>N</i> Left of cutoff	85741	85741	85741	85741
<i>N</i> right of cutoff	67123	67123	67123	67123
BW left	5.040	5.040	5.040	5.040
BW right	10.60	10.60	10.60	10.60

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 8 Appendix. Equivalent to Table 9 of the main paper. Full sample with first observation as treatment assignment. Column 1 (speeding) is a dummy that takes the value of 1 if the vehicle had at least one speeding, and 0 otherwise. Column 2 (Times speeding) is a continuous variable, the total of times a vehicle had at least one speeding. Column 3 (Speeding>10) is a dummy that takes the value of 1 if the vehicle had at least one serious

speeding (above 60 km/h), and 0 otherwise. Column 4 (Times speeding>10) is a continuous variable, the total of times a vehicle had at least one serious speeding.

My main approach to including vehicle that pass more than once is different. Since the problem is how to assign to treatment, this problem is not present when a vehicle passes always below or above the limit. In the following vehicles that fulfill this condition are also used and their assigned speed is the average of all the measures of speed. The assigned date is the first time it passed above the legal limit under a Speed Camera. Table 9 presents those results. Note that since the conditions is more strict, there is a smaller increase on sample size, of only 6% (from 136553 to 145080). These observations also tend to speed more.

Table 9: Using also vehicles always on the same side of the cutoff

	(1) Speeding dummy	(2) Speeding continuous	(3) Speeding >10 dummy	(4) Speeding >10 continuous
Bias-corrected	-0.158*** (0.0256)	-0.235*** (0.0571)	-0.0454*** (0.0141)	-0.0537** (0.0219)
Mean of not ticketed	0.261	0.446	0.0603	0.0787
N	145080	145080	145080	145080
N Left of cutoff	81403	81403	81403	81403
N Right of cutoff	63677	63677	63677	63677
BW left	5.040	5.040	5.040	5.040
BW right	10.60	10.60	10.60	10.60

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 9 Appendix. Equivalent to Table 9 of the main paper. Full sample with vehicle that pass more than once at the same side of the cutoff. Column 1 (speeding) is a dummy that takes the value of 1 if the vehicle had at least one speeding, and 0 otherwise. Column 2 (Times speeding) is a continuous variable, the total of times a vehicle had at least one speeding. Column 3 (Speeding>10) is a dummy that takes the value of 1 if the vehicle had at least one serious speeding (above 60 km/h), and 0 otherwise. Column 4 (Times speeding>10) is a continuous variable, the total of times a vehicle had at least one serious speeding.

All other results (not reported) for these augmented samples are consistent for the findings of the main body of the paper. The only slight difference has to do with the relative price coefficients for the matched sample when using first observation as treatment assignment. Even if the effect is still concentrated on higher quintiles, the effect is higher in magnitude for quintile 3 than quintile 5. This even if the most

important finding, the absence of an effect on lower quintiles of vehicle price, is consistent across all estimations, no matter how the sample is selected.